Bounds on Round Skipping Attacks on the SPN Hash functions

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1 Introduction

Zero-knowledge proof systems—SNARKs, STARKs, and Bulletproofs—have become core infrastructure for privacy-preserving computation. Their inner loops are dominated by hash computations that must be expressed natively over prime fields with as few multiplications as possible. The Poseidon family of hash functions [1, 2], built on the Substitution-Permutation Network (SPN) design, is the dominant answer: deployed in Filecoin, Mina, Plonky3, and many other production systems, it achieves the best known trade-off between algebraic complexity and security.

An SPN permutation operates on a state $W \in \mathbb{F}_p^t$ by alternating a non-linear S-box layer $\mathcal{S}$ (applying the power map $x \mapsto x^d$ to each element) with a linear diffusion layer $\mathcal{L}$ (multiplication by a $t \times t$ matrix M). Security is measured primarily through the *Constrained-Input Constrained-Output* (CICO) problem: given partial constraints on the input and output of the permutation, find a satisfying assignment. CICO hardness underlies collision resistance and preimage resistance, and concretizes the security of the Poseidon Initiative bounty challenges [5].

Round-skipping attacks reduce the CICO problem by algebraically bypassing a prefix of rounds. Merz and Rodríguez García [3] recently showed that the structured linear layers of Poseidon2(b)—optimized for hardware efficiency but not MDS—admit skipping attacks that bypass up to $\approx t/4$ full rounds by spending roughly four degrees of freedom per skipped round. This raises a fundamental open question: *how many rounds can be skipped when the linear layer is an MDS matrix?* Poseidon1 uses MDS matrices (via the Cauchy or circulant construction, achieving branch number $b = t + 1$), so this question directly bounds the attack surface of the Poseidon Initiative bounty instances.

Our Contributions

- **Skipping trail framework (Section 3).** We introduce *skipping trails*: truncated differential trails satisfying an algebraic compatibility condition that forces the ratio of active state element pairs to propagate linearly through the S-box layer. A valid m-round skipping trail certifies that a structured family of inputs maps through m rounds to a predictable output set, enabling the adversary to reduce a CICO instance on the full permutation to one on the remaining $R_F - m$ full rounds and R_P partial rounds.

- **Tight bounds on round skipping (Section 4).** We prove that an m-round skipping trail can exist only if the branch number b of $\mathcal{L}$ satisfies

$$b \le \frac{4t}{m} \quad (m \text{ even}), \qquad b \le \frac{4t-2}{m-1} \quad (m \text{ odd}).$$

  For MDS matrices ($b = t + 1$), this rules out skipping $m \ge 4$ rounds for all $t \ge 1$. For $m = 3$, the bound gives $b \le 2t - 1$, which MDS satisfies, so three-round skipping is not excluded.

- **Concrete trails for Poseidon1-KoalaBear (Section ??).** For the canonical bounty instance ($p = 2^{31} - 2^{24} + 1$, $d = 3$, $t = 16$, $R_F = 8$, $R_P = 20$), we exhibit explicit 2-round skipping trails for both the circulant and Cauchy MDS constructions. We further search for 3-round skipping trails; a positive result would constitute the first example of three-round MDS round skipping.
- **Complexity analysis and implementation (Section ??).** We quantify the CICO cost reduction as a function of the trail width and R_P. We provide an implementation that finds skipping trails for concrete Poseidon1-KoalaBear matrices and demonstrates the round-skipping attack on the bounty instances.

Scope. We analyze skipping within the full-round layer only ($m \leq R_F$). Extending skipping into the partial-round layer is qualitatively different and is left for future work.

Organization. Section 2 covers SPN notation, Poseidon1, MDS matrices, and the CICO problem. Section 3 defines truncated differential trails and skipping trails. Section 4 proves the main bounds. Section **??** presents concrete trails for Poseidon1-KoalaBear. Section **??** describes the implementation and complexity analysis.

1.1 Related Work

Poseidon hash functions. Poseidon1 [1] (Grassi, Khovratovich, Rechberger, Roy, Schofnegger, 2021) is an SPN hash function over $\mathbb{F}_p$ with R_F *full rounds* (all t S-boxes active) flanking R_P *partial rounds* (one S-box active), and an MDS linear layer with branch number $b = t + 1$. Poseidon2 [2] replaces the MDS matrix with a structured matrix achieving better circuit efficiency; this matrix is not MDS in general, and its reduced diffusion is the structural weakness exploited in [3].

Round-skipping attacks. The idea of bypassing SPN rounds algebraically has roots in the *rebound attack* of Mendel, Rechberger, Schläffer, and Thomsen [4] on AES-based hash functions. In the rebound attack, the adversary satisfies differential constraints through the middle rounds by working inward from both sides simultaneously; the adversary controls the degrees of freedom through the linear (MDS) layer and the non-linear S-box. Merz and Rodríguez García [3] adapted this idea to Poseidon2 by identifying truncated differentials with a ratio-preserving property through the power-map S-box. Their attack skips up to $\approx t/4$ rounds and yields the first algebraic collision attack that outperforms the corresponding preimage attack on Poseidon2. They explicitly leave open the question of how far round skipping can go for MDS-based SPNs; our paper answers this question.

Algebraic attacks. Algebraic attacks on Poseidon-type primitives include Gröbner-basis methods, linearization, and interpolation. These complement round-skipping attacks: our results show that, for MDS matrices, at most 3 full rounds can be skipped (with $m \geq 4$ being provably impossible), bounding the scope of the round-reduction phase in any hybrid attack.

MDS matrices and branch numbers. The branch number $b = \min_{x \neq 0}(\mathrm{wt}(x) + \mathrm{wt}(Mx))$ quantifies the diffusion of a linear layer, where wt denotes Hamming weight. MDS matrices achieve $b = t + 1$, the maximum by the Singleton bound. The Cauchy construction $M[i][j] = (x_i - y_j)^{-1}$ yields an MDS matrix over any prime field with $p > 2t$; circulant MDS matrices, used in Plonky3

and the Poseidon Initiative bounty instances, achieve the same branch number with a more regular structure. The distinction between the two constructions becomes relevant in Section **??**.

2 Preliminaries and Notation

2.1 SPN Permutations

Let $\mathbb{F} = \mathbb{F}_p$ be a prime field and $t \geq 2$ an integer. An *SPN permutation* $\mathcal{P}$ operates on the state $W \in \mathbb{F}^t$ as a composition of R rounds. Each round i applies three operations in sequence:

1. $\mathcal{L}$ — a linear transformation that multiplies the state by a fixed matrix $M \in \mathbb{F}^{t \times t}$;
2. $\mathcal{S}$ — a non-linear substitution layer that applies the power map $x \mapsto x^d$ coordinate-wise;
3. $\mathcal{C}_i$ — a constant-addition layer that adds a round-specific vector of constants to the state.

We write the i-th round as

$$W_i^{(A)} \xrightarrow{\mathcal{L}} W_i^{(B)} \xrightarrow{\mathcal{S}} W_i^{(C)} \xrightarrow{\mathcal{C}_i} W_{i+1}^{(A)}.$$

Here $W_i^{(A)}$ is the state at the beginning of round i (after constants of round $i-1$), $W_i^{(B)}$ is the state after the linear layer, and $W_i^{(C)}$ is the state after the S-box layer.

2.2 The CICO Problem

The primary security notion for SPN-based hash functions is the *Constrained-Input Constrained-Output* (CICO) problem.

Definition 1 (CICO)**.** *Given an SPN permutation $\mathcal{P} : \mathbb{F}^t \to \mathbb{F}^t$ and affine constraints*

$$\mathcal{C}_{\text{in}} \subseteq \mathbb{F}^t, \quad \mathcal{C}_{\text{out}} \subseteq \mathbb{F}^t,$$

find $W \in \mathcal{C}_{\text{in}}$ such that $\mathcal{P}(W) \in \mathcal{C}_{\text{out}}$, or determine that no such W exists.

The CICO problem underlies collision and preimage resistance of Poseidon-based hash functions. Concretely, fixing the capacity elements of the input and output yields a collision; fixing the output fully yields a preimage. The Poseidon Initiative bounty challenges [5] are cast as CICO instances.

2.3 Poseidon1

Poseidon1 [1] is an SPN permutation over $\mathbb{F}_p^t$ with a *mixed-round* structure: it consists of R_F *full rounds*, in which the S-box layer is applied to all t state elements, flanking R_P *partial rounds*, in which only one designated element is cubed. Concretely, the round schedule is $R_F/2$ full rounds, followed by R_P partial rounds, followed by $R_F/2$ full rounds; the total number of rounds is $R = R_F + R_P$.

The canonical bounty instance over the KoalaBear field uses the parameters

$$p = 2^{31} - 2^{24} + 1, \quad d = 3, \quad t = 16, \quad R_F = 8, \quad R_P = 20.$$

Scope. This paper analyzes round skipping within the full-round layers only ($m \leq R_F$). Extending skipping into the partial-round layer is qualitatively different and is left for future work.

2.4 MDS Matrices and Branch Number

Definition 2 (Branch number)**.** *The* branch number *of a linear map $\mathcal{L}$ with matrix M is*

$$b(M) = \min_{x \neq 0} \big(\mathrm{wt}(x) + \mathrm{wt}(Mx) \big),$$

where $\mathrm{wt}(v)$ *denotes the Hamming weight (number of non-zero coordinates) of $v \in \mathbb{F}^t$.*

Definition 3 (MDS matrix)**.** *A matrix $M \in \mathbb{F}^{t \times t}$ is* MDS *(Maximum Distance Separable) if $b(M) = t + 1$, the maximum allowed by the Singleton bound.*

An equivalent characterization: M is MDS if and only if every square submatrix of M is non-singular.

Two MDS constructions appear in deployed instances of Poseidon1.

- **Cauchy MDS.** For distinct elements $x_1, \ldots, x_t, y_1, \ldots, y_t \in \mathbb{F}_p$ with $x_i \neq y_j$ for all i, j, the matrix $M[i][j] = (x_i - y_j)^{-1}$ is MDS over any prime field $p > 2t$.
- **Circulant MDS.** A circulant matrix M defined by its first row $(c_0, c_1, \ldots, c_{t-1})$ with $M[i][j] = c_{(j-i) \bmod t}$ is MDS for specific choices of $(c_0, \ldots, c_{t-1})$ depending on p and t. The Poseidon Initiative bounty instances and Plonky3 use circulant MDS matrices.

Both constructions achieve $b = t + 1$; the distinction becomes relevant in Section **??**.

3 Truncated Differentials and Skipping Trails

Definition 4. *A set $\mathcal{W} \subseteq \mathbb{F}^t$* satisfies the matrix differential $\mathbf{A}$*, centered at $W_0 \in \mathbb{F}^t$, if*

$$\mathcal{W} = \big\{ W_0 + \mathbf{A} \cdot \mathbf{X} : \mathbf{X} \in \mathbb{F}^t \big\}.$$

We call $\mathcal{W}$ a differential coset *of $\mathbf{A}$.*

Definition 5. *A* truncated differential *is a pair of matrices $(\mathbf{A}, \mathbf{B}) \in (\mathbb{F}^{t \times t})^2$ such that*

1. *$\mathbf{A} \xrightarrow{\mathcal{L}} \mathbf{B}$, i.e. for every differential coset $\mathcal{W}$ of $\mathbf{A}$, its image $\mathcal{L}(\mathcal{W})$ is a differential coset of $\mathbf{B}$;*
2. *Matrix $\mathbf{B}$ is diagonal.*

The indices j with a non-zero j-th row of $\mathbf{A}$ (equivalently, non-zero j-th diagonal entry of $\mathbf{B}$) are called the active positions*.*

Definition 6. *A diagonal matrix $\mathbf{B}$ is $\mathcal{S}$-*compatible *with a matrix $\mathbf{A}'$ if there exist differential cosets $\mathcal{W}^{(B)}$ of $\mathbf{B}$ and $\mathcal{W}^{(A)}$ of $\mathbf{A}'$ such that*

$$\mathcal{S}\Big(\mathcal{W}^{(B)}\Big) = \mathcal{W}^{(A)},$$

where $\mathcal{S}$ applies the map $x \mapsto x^d$ coordinatewise. (Round constants shift the center of $\mathcal{W}^{(A)}$ and do not affect $\mathcal{S}$-compatibility.)

Definition 7. *A m-round* truncated differential trail *is a sequence of truncated differentials $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^m$ such that for every $i = 1, \ldots, m-1$, $\mathbf{B}_i$ is $\mathcal{S}$-compatible with $\mathbf{A}_{i+1}$.*

Definition 8. *A m-round* skipping trail *is a $m+1$-round truncated differential trail $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{m+1}$, where for every i from 1 to m, there exists a decomposition the active elements of $\mathbf{B}_i$ into two equally-sized groups L_i and R_i, such that the trail, now in the form $\{(\mathbf{A}_{i,L}, \mathbf{A}_{i,R}, \mathbf{B}_{i,L}, \mathbf{B}_{i,R})\}_{i=1}^{m+1}$ satisfies the following conditions:*

$$\frac{\mathbf{B}_{i,R}^d}{\mathbf{B}_{i,L}^d} \circ \mathbf{A}_{i+1,L} = \mathbf{A}_{i+1,R}$$

where the division of diagonal matrices is defined as the element-wise division of the diagonal elements.

With an m-round skipping trail, we can algebraically skip the first m rounds of the SPN structure, and directly jump to the state after the m-th round. The following theorem formalizes this intuition.

4 Theoretical Bounds on Round Skipping

Theorem 1. *Suppose there is a m-round skipping trail such that for some subset I of state elements we have $A_1[I, *] = \mathbf{0}$, and the total number of active elements is $k < 2t$. If there exists a sequence of states $\{W_i^{A,B,C}\}_{i=1}^{m+1}$ such that for every i from 1 to m we have*

$$\mathbf{B}_{i,L} \cdot W_{i,R}^{(B)} = \mathbf{B}_{i,R} \cdot W_{i,L}^{(B)} \tag{1}$$

Then the truncated differential trail $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{m+1}$ is a valid truncated differential trail, and the probability of the truncated differential is at least $p^{-k/2}$. Moreover, it holds

$$\{\mathcal{P}^m(W_1^A + \mathbf{A}_1 \cdot \mathbf{X})\}_{\mathbf{X} \in \mathbb{F}^t} = \{W_{m+1}^C + \mathbf{B}_{m+1} \cdot \mathbf{Y}\}_{\mathbf{Y} \in \mathbb{F}^t} \tag{2}$$

Proof. TODO (just apply the definition of skipping trail and the condition 1 to get 2). □

Corollary 1 (CICO degree reduction)**.** *Let $m \leq R_F$ and suppose there exists an m-round skipping trail $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{m+1}$. Suppose the CICO input constraint defines a differential coset of $\mathbf{A}_1$, and let the composition of the remaining $s = R - m$ rounds of $\mathcal{P}$, viewed as a function of the state at the beginning of round $m+1$, be expressed by polynomials of degree at most D over $\mathbb{F}$. Then the CICO problem reduces to a system of polynomial equations of degree at most D.*

Proof. By Theorem 1, the image of the input differential coset under the first m rounds is the differential coset $\{W_{m+1}^C + \mathbf{B}_{m+1} \cdot \mathbf{Y}\}_{\mathbf{Y} \in \mathbb{F}^t}$. The remaining s rounds are a polynomial map of degree D applied to this affine family; substituting the parameterization by $\mathbf{Y}$ turns the CICO output constraints into polynomial equations in $\mathbf{Y}$ of degree at most D. (The number of free variables equals the rank of $\mathbf{B}_{m+1}$, i.e. the number of active positions of $\mathbf{A}_{m+1}$.) □

Proposition 1. *A sequence of states satisfying an m-round skipping trail exists only if the branch number b of the linear transformation $\mathcal{L}$ satisfies the following condition:*

$$\Big\{ b \leq \tfrac{4t}{m}, \text{ for } m \text{ even} \tag{3}$$

Proof. Suppose there is a sequence of states $\{W_i^{A,B,C}\}_{i=1}^{m+1}$ satisfying the condition 1. Then for every i from 1 to m we have

$$\mathbf{B}_{i,L} \cdot W_{i,R}^{(B)} = \mathbf{B}_{i,R} \cdot W_{i,L}^{(B)} \tag{4}$$

Denote the number of active elements in $\mathbf{B}_i$ by k_i. A satisfying sequence of states can exist only if

$$\sum_i k_i/2 \leq t.$$

On the other hand, any pair of states satisfying a truncated differential also satisfies a regular differential trail with at most the same number of active elements. Therefore, we must have $k_i + k_{i+1} \geq b$ for every i from 1 to $m-1$. Combining these two inequalities we get for even m:

$$t \geq \sum_i k_i/2 \geq \frac{m}{4} b$$

□

Proposition 2. *If the linear transformation $\mathcal{L}$ is MDS, then a 4-round skipping trail cannot exist.*

Proof. Follows from the fact that for an MDS matrix, the branch number is maximal $t+1$, and thus the condition for a 4-round skipping trail would be $t+1 \leq \frac{4t}{4}$, which is not satisfied for any $t \geq 1$. □

Remark 1. *Four rounds of an MDS-based SPN structure can potentially be skipped if the differential for the first round is relaxed.*

References

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